Graph Topologies and Incentive Stability in Multi-Agent Code Generation

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Motivation

Do complex graph structures inherently discourage high-effort verification from individual agents due to varying systemic incentives?

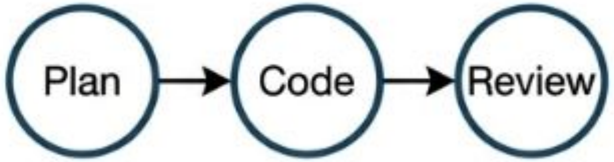
- Multi-agent systems introduce rapidly compounding token and latency costs
- Lyu et al. (2025) show that agent failures trace directly to inter-agent miscommunication, implicating topology affect both accuracy and robustness
- We must isolate the true hidden variable: **Interaction Topology**

Background

Baseline

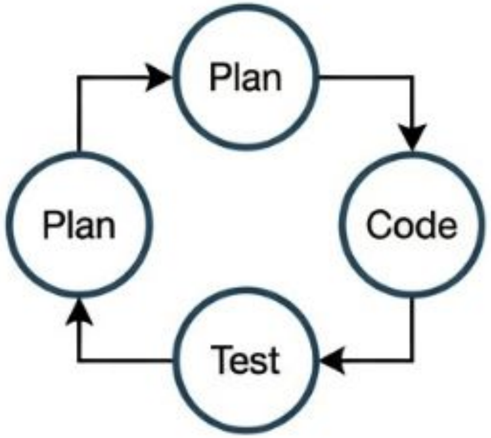
Single LLM zero-shot prompts (early GPT-3). Low functional correctness

Orchestrated Pipelines



Frameworks like MetaGPT and AutoGen formalize roles to prevent cascading errors

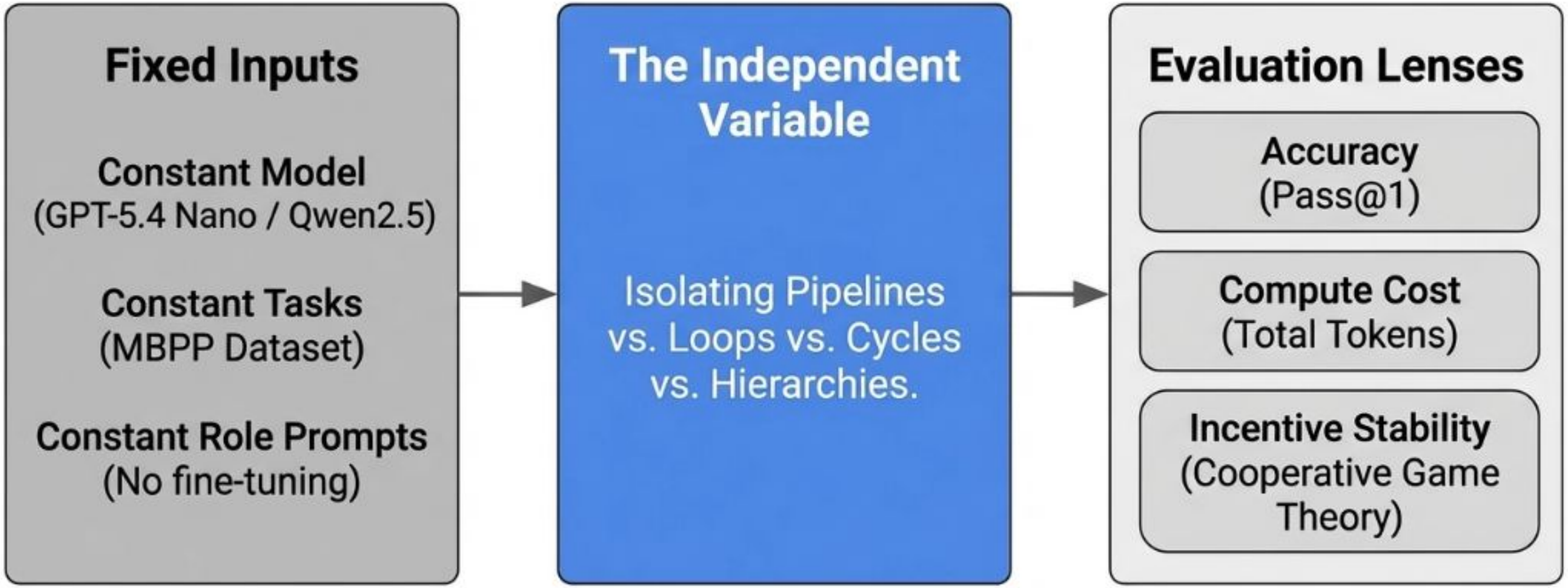
Closed-Loop state-of-the-art



Reflexion, AgentCoder, MapCoder. Iterative testing maximizes pass rates

Gap: Current state-of-the-art evaluates systems as holistic black box. → We lack isolated, controlled analysis of how the Interaction Topology itself drives performance and token economics

Overview: Our Approach



Game Theory Formulation

Analyzing multi-agent coding as a cooperative game where completion is a shared payoff

Overall Utility for Agent i

Compute cost scalar

$$U_i = V \cdot 1_{[Y=pass]} - \lambda \cdot T_i$$

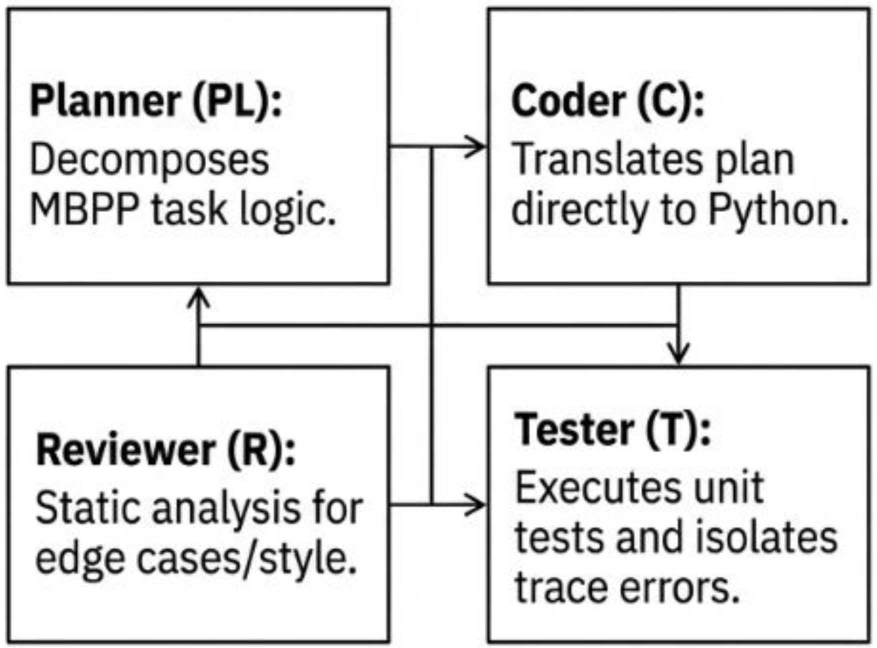
The inherent value of task success (shared payoff)

Binary execution outcome (1 if MBPP unit tests pass, 0 if fail)

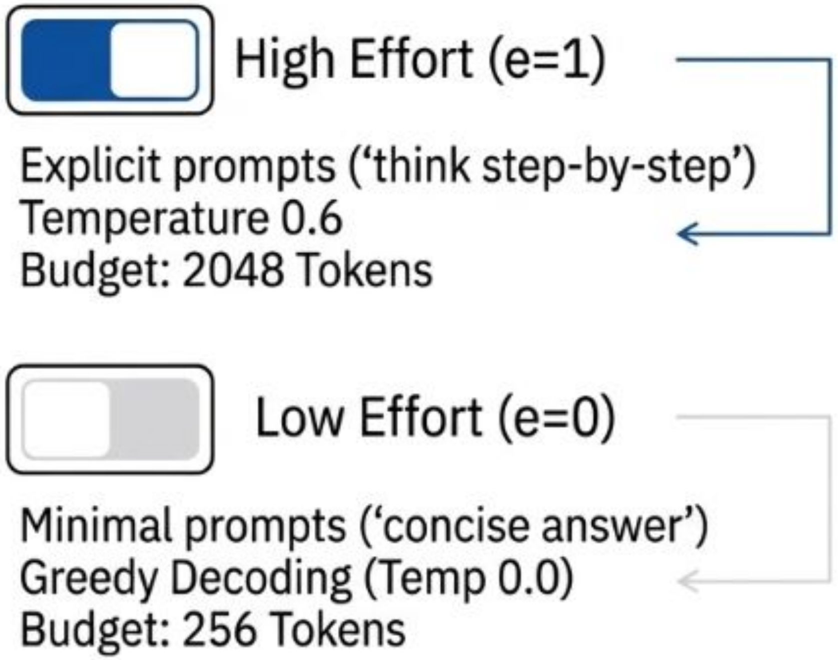
Token expenditure (effort cost) by Agent i

Operational Framework: Agent Roles and Effort Constraints

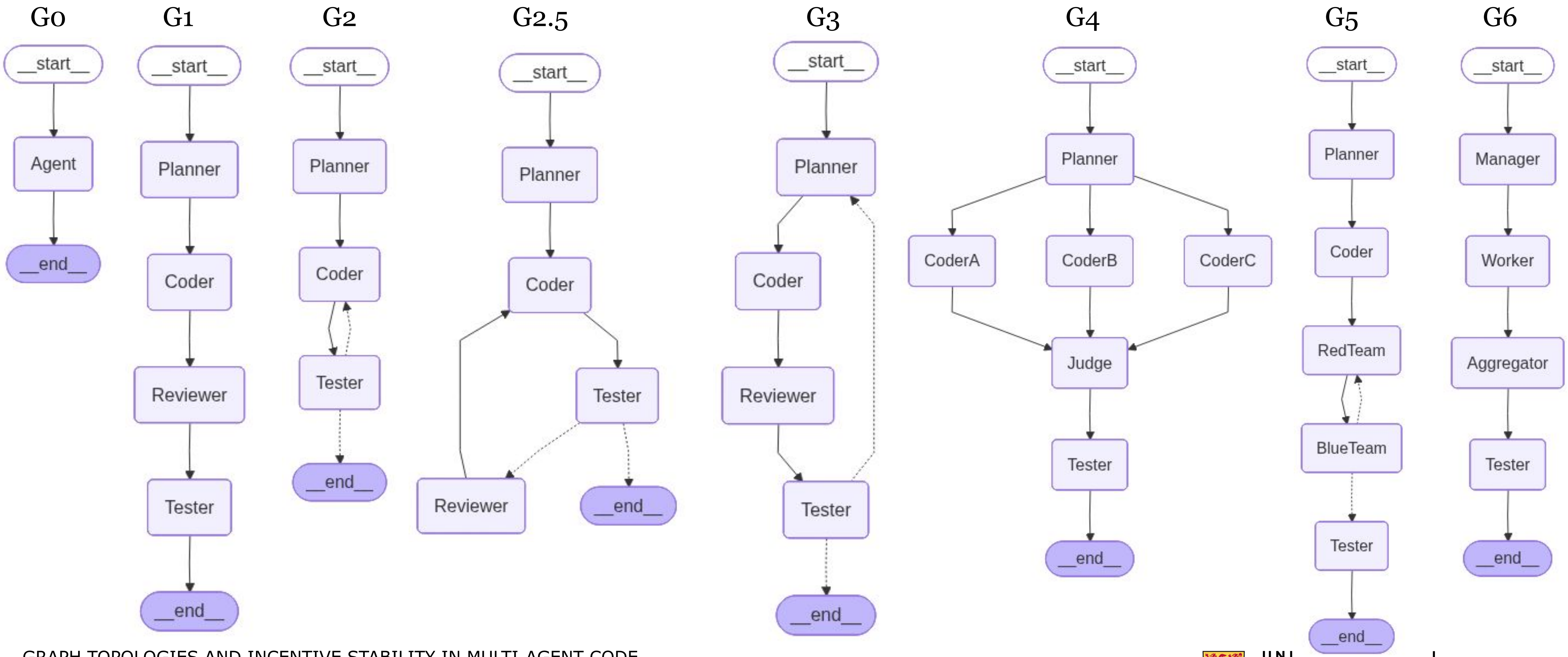
The Agent Nodes



Effort ($e \in \{0, 1\}$)



Graph Topologies Overview



Implementation

Dataset

- 50 targeted coding tasks sourced from the MBPP (Mostly Basic Python Problems) Benchmark.

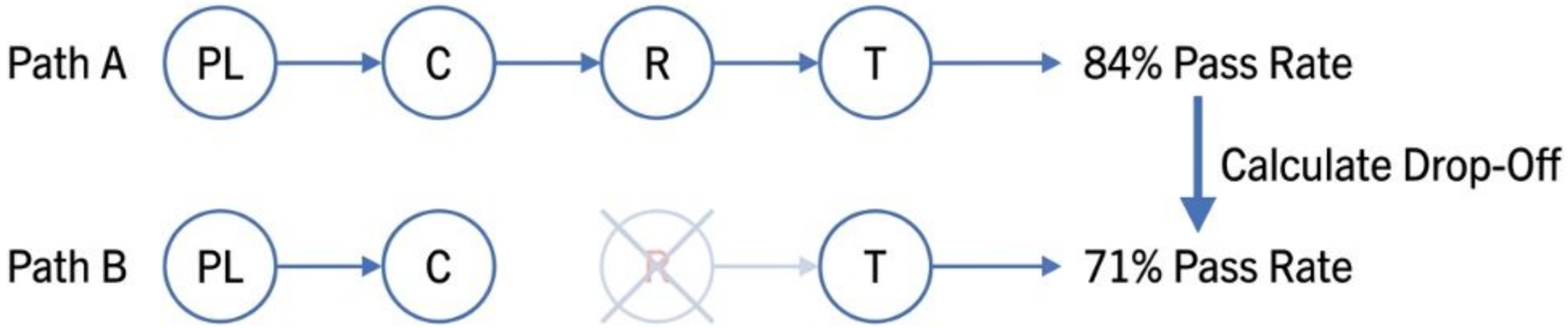
Orchestration

- Graph Topologies implemented using LangGraph. These topologies define how information flows between specialized agent nodes
- Telemetry callback handlers strictly trace prompt/completion tokes to calculate token expenditure

Evaluation: Shapley Values

How much credit does a specific agent (e.g., Reviewer) truly deserve for a successful execution pass?

$$\phi_i(g) = \mathbb{E}_\pi \left[v_g(S_\pi^i \cup \{i\}) - v_g(S_\pi^i) \right]$$

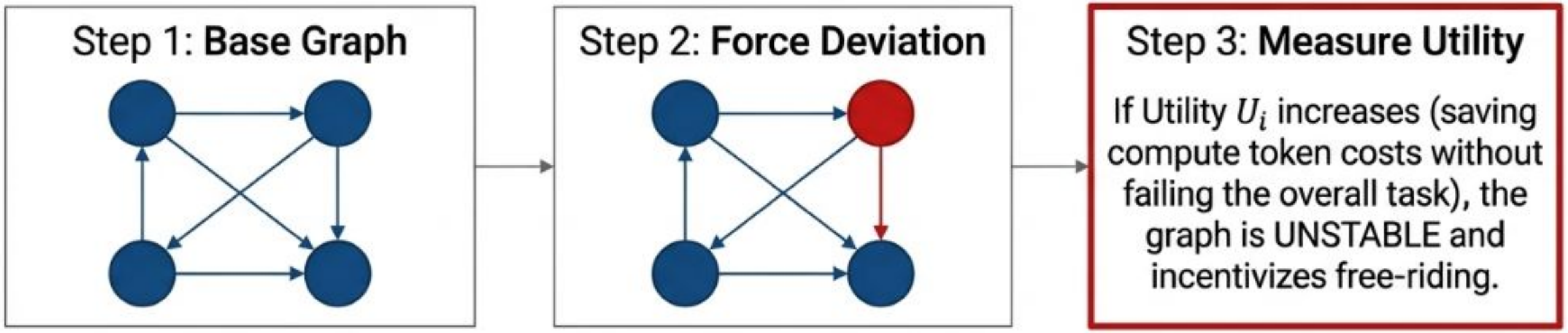


Evaluation: Nash Equilibrium Test

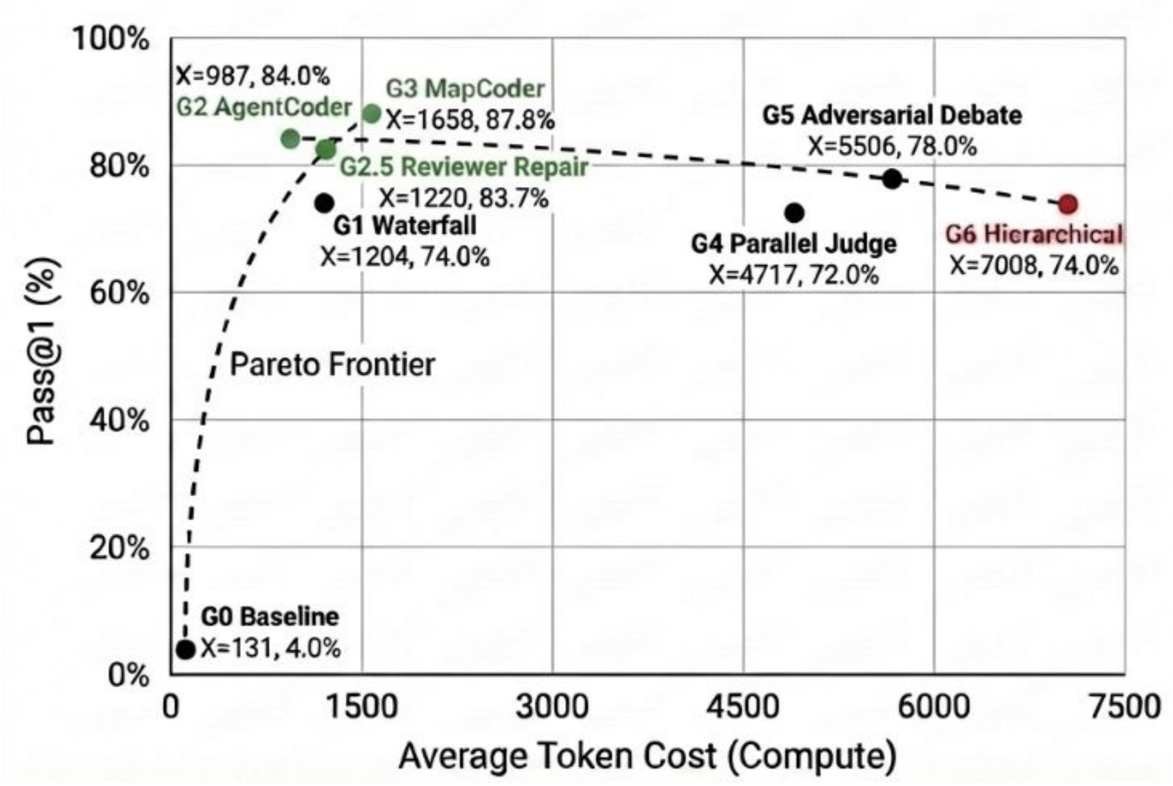
Does the architectural graph topology encourage agents to slack off?

A topology is stable if NO agent **benefits** from **unilaterally** dropping to **low effort** (ε-Nash Equilibrium):

$$U_i(e_i = 1, e^*_{-i}) \geq U_i(e_i = 0, e^*_{-i}) - \varepsilon$$



Results



- Cyclic topologies (G2, G3) vastly outperform other complex parallel graphs (G4, G6) in both accuracy and token cost

Conclusion

Discussion & Limitation

Extend the evaluation to **broader benchmarks** and more realistic software engineering workflows.

Q&A

Thank You

Motivation

Do complex graph structures inherently discourage high-effort verification from individual agents due to varying systemic incentives?

- Task decomposition reduces errors but explodes costs
- We must understand the hidden variable: Interaction Topology